# Mobile Surgical Humanoid: System Datasheet

**Part no.:** H2-SURGICAL-1.0 (hypothetical 2030)
**Revision:** A
**Class:** investigational

Kevin Kawchak, CEO ChemicalQDevice | 10.5281/zenodo.xxxxxxxx | June 5, 2026

**Abstract.**

This datasheet template summarizes a mobile surgical humanoid whose robot-patient interaction code is verified before generation, with specifications, interfaces, and operating envelope.

*Disclaimer: This work is independent and not endorsed or sponsored by trial sponsors, FDA, CROs, sites, IRBs, regulators, or medical societies; and was generated using Artificial Intelligence.*

### Index terms

## 1 System overview

The system is a mobile surgical humanoid with a sensing-planning-actuation stack gated by verify-before-generate. It is a clearly labeled hypothetical 2030 platform.

*Figure placeholder (`Images/block-diagram.png`). The float, caption, and `\label` mirror the VVUQ-02 figures; replace the box with `\includegraphics`.*

**Figure 1:** Figure 1. Placeholder block diagram of the humanoid system.

## 2 Specifications

| Parameter | Value | Notes |
|---|---|---|
| sensing cadence | 10 Hz | sensor-to-xyz |
| planning cadence | 1 Hz | LLM planner |
| contact-force budget | 15 / 5 N | per ISO/TS 15066 direction |

**Table 1:** Table 1. Key specifications (modeled on VVUQ-05 Table 1).

## 3 Interfaces

External interfaces.

`sensor bus` streams to the sensor-to-xyz stage at 10 Hz

`planner link` receives 1 Hz plans from the on-premises LLM

`safety channel` carries the e-stop and clearance signals

## 4 Operating envelope

The system operates only within its readiness gate and contact-force budget; outside the envelope the gate forces a safe state before any motion is generated.

### Rights and Permissions



### Cite This Datasheet

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### Data Availability

Supporting records are openly available in the `kevinkawchak/cancer-automated` and `kevinkawchak/physical-ai-oncology-trials` GitHub repositories and are archived on Zenodo. All data sets are synthetic or illustrative and reproducible from a deterministic seed; no patient-identifiable information is included.

## References

[1] K. Kawchak, "VVUQ Processing Priority over Code Generation: Method and Pipeline." `https://doi.org/10.5281/zenodo.20372501`, 2026. `https://doi.org/10.5281/zenodo.20372501`.

[2] K. Kawchak, "Mobile Pancreatic Cancer Unitree H2 Surgical Humanoid with Priority VVUQ." `https://doi.org/10.5281/zenodo.20421754`, 2026. `https://doi.org/10.5281/zenodo.20421754`.

[3] K. Kawchak, "Verification Before Generation in Physical AI Oncology Trials Act of 2026 (H. R. 9510)." `https://doi.org/10.5281/zenodo.20535429`, 2026. `https://doi.org/10.5281/zenodo.20535429`.

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[6] L. Lamport, *LaTeX: A Document Preparation System*. Addison-Wesley, 2nd ed., 1994.